

SAKSHAM UPADHYAY

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Executive Summary

I am an Electronics and Communication Engineering student who is always been curious about applying technology to solve real-world problems that directly impact human lives. This curiosity has evolved into a strong interest in advanced engineering systems, research, and innovation-driven environments. I am motivated by continuous learning, hands-on exploration, and contributing to meaningful, technology-driven solutions.

Education

VIT Bhopal University
Bachelor of Technology
Major in Electronics and Communication Engineering

Bhopal, MP
Cumulative: 8.36/10
Graduating in May 2026

Relevant Coursework : OOPS, DBMS

Technical Skills

Languages: Java, Python, SQL

Libraries/Frameworks: LangChain, CrewAI, TensorFlow

AI/ML Expertise: Machine Learning, Generative AI, Natural Language Processing

Experience

Pawspal Connect

December 2025 - Present

AI Intern

- Contributed to **EMR system development** by implementing AI-based data extraction and processing using **Python** and **LangChain**.
- Integrated **Google Gemini LLM** to enable natural language querying and document summarization within EMR workflows.
- Developed **RESTful APIs** using FastAPI to support LLM integration and scalable backend automation.

Projects

CodeCritic AI | Python, LangChain, Google Gemini API

(July 2025 – August 2025)

- Automated code analysis for **10+ programming languages**, reducing manual review time by **80%**.
- Designed **dual-mode LLM engine**, combining rapid feedback and deep multi-tool analysis.
- Reduced developer workload by **15 hours per week** with automated categorization into **6 priority sections**.

DocGenie | Python, LangChain, Google Gemini API

(February 2025 – March 2025)

- Built a **multilingual AI PDF chatbot** capable of answering queries from **100+ page documents in 4+ languages**.
- Integrated **semantic search with FAISS** over **1,000+ text chunks**, boosting retrieval accuracy by **30%**.
- Added **voice-based query handling** (gTTS), improving accessibility for **100% of users**.

Deep Fake Audio Detection | Python, Deep Learning, Audio Processing

(November 2024 – December 2024)

- Constructed a CRNN model achieving 73%+ accuracy in detecting audio deepfakes with 3 convolutional layers and a 2-layer bidirectional LSTM, trained on 1,000+ real/fake audio samples.
- Optimized audio preprocessing by standardizing variable-length inputs to fixed 200-frame spectrograms (16 kHz SR, 1024 n FFT), reducing inference latency by 30% through GPU-accelerated batch processing (16 samples/batch).
- Reduced overfitting by 15% via dropout (0.3) and adaptive learning rate scheduling (patience=2), while maintaining 90%+ validation accuracy across 10 training epochs.

Certifications

Generative AI Certification - IBM (April 2025)

Java Programming Certification - SkillVertex (March 2023)